

ROMAN DOMINATION AND ITS VARIANTS IN UNIT DISK GRAPHS*

WEIPING SHANG^{†,§}, XIUMEI WANG[†] and XIAODONG HU[‡]

[†]*Department of Mathematics, Zhengzhou University
 Zhengzhou 450001, China*

[‡]*Institute of Applied Mathematics
 Chinese Academy of Sciences
 Beijing 100080, China*

[§]*shangwp@zzu.edu.cn*

Accepted 1 February 2010

Unit disk graphs are the intersection graphs of equal sized disks in the plane, they are widely used as a mathematical model for wireless ad-hoc networks and some problems in computational geometry. In this paper we first show that Roman dominating set and connected Roman dominating set problems in unit disk graphs are NP-complete, and then present two approximation algorithms for these problems.

Keywords: Dominating; approximation algorithm; unit disk graph.

Mathematical Subject Classification 2000: 68W25

1. Introduction

Let $G = (V, E)$ be a simple and undirected graph. A set $D \subseteq V$ is a dominating set if every vertex in $V \setminus D$ is adjacent to at least one vertex in D . The domination number, denoted $\gamma(G)$, is the minimal cardinality of a dominating set in G . The minimum dominating set problem is to compute a dominating set of minimal cardinality for any given graph G . A Roman dominating function [2] of a graph G is defined as a function $f : V(G) \rightarrow \{0, 1, 2\}$ satisfying the condition that every vertex u with $f(u) = 0$ is adjacent to at least one vertex v with $f(v) = 2$. The weight of a Roman dominating function is $f(V) = \sum_{v \in V} f(v)$. The Roman domination number, denoted $\gamma_R(G)$, is the minimum weight of Roman dominating functions in G . The minimum Roman dominating set problem is to compute a minimum weight dominating function for any given graph G .

*This work was supported by the National Natural Science Foundation of China under Grant No. 10901144.

[§]Corresponding author.

Minimum Roman dominating set problem is introduced in [2] as one of variants of classical dominating set problem. It comes with a nice story: The Roman Empire needs to defence itself by positioning some legions on the various parts of the Empire in such a way that either (1) a specific region v is also the location of at least one legion or (2) one region u neighboring v has two legions, so that u can afford sending off one army to the region v (in case of an attack) without losing self-defence capabilities. Since it is expensive to maintain a legion at a location, the Emperor would like to station as few legions as possible, while still defending the Roman Empire. Roman dominating set problem has not only historical but also mathematical interests, particularly in the field of server placements [8].

In reality, it is desired that dispatching a legion from one location to another location should not take too much time; In particular, it is desirable that a legion from one location could be dispatched to another location if the Euclidean distance between the two corresponding points is no more than a specified bound. Unit disk graphs satisfy such a property. A undirected graph $G = (V, E)$ is a unit disk graph (UDG) if its vertices can be put in one-to-one correspondence with disks of equal radius in the plane in such a way that two vertices are joined by an edge if and only if the corresponding disks intersect. In other words, for n points in the plane, form a graph with n vertices corresponding to the n points, and an edge between two vertices if and only if the Euclidean distance between the two corresponding points is at most some specifies bound d . Grid graphs compose a subclass of unit disk graphs under such an intersection model that all the disks have centers with integer coordinates and radius $1/2$.

It has been shown in [1] that many graph related problems in unit disks graphs are NP-complete, including the independent set problem, the dominating set and connected dominating set problems; In fact, the dominating set and connected dominating set problems are NP-complete even for grid graphs. Roman dominating set problem in general graphs is proved to be NP-hard in [3], $(2 + 2 \ln n)$ -approximation algorithms and a PTAS are proposed in [8] for general graphs and for minimum planar Roman dominating set problem, respectively. Liedloff *et al.* [6] show that the Roman domination number of an interval graph can be computed in linear time.

In this paper, we consider the Roman dominating set and connected Roman dominating set problems in unit disk graphs. We will first give some preliminaries about the problem in Sec. 2, and then we will prove that these problems in unit disk graphs are NP-complete in Sec. 3, after that we will propose two approximation algorithms in Sec. 4. In Sec. 5 we conclude the paper.

2. Preliminaries

Let $G = (V, E)$ be a graph. For any vertex $v \in V$, the neighborhood of v is defined by $N_G(v) \equiv \{u \in V(G) : uv \in E(G)\}$. The degree $d_G(v)$ of a vertices v in G is the number of edges incident with v . Let V' be a subset of V , denote by $G[V']$ the induced graph of vertex set V' .

A subset $U \subseteq V$ is called an *independent set* (IS) of G if all vertices in U are pairwise non-adjacent, and it is further called a *maximal independent set* (MIS) if each vertex in $V \setminus U$ is adjacent to at least one vertex in U . Clearly, an MIS I of G is also a dominating set of G . Thus any minimum dominating set D of G has size no more than that of an MIS I . In addition, a subset S of V is called a 2-packing if $N(u) \cap N(v) = \emptyset$ for every pair of vertices $u, v \in S$.

Some useful facts on Roman dominating functions were proved in [2], which will be used in our following discussions.

Proposition 2.1. *For any graph G , $\gamma(G) \leq \gamma_R(G) \leq 2\gamma(G)$.*

Proposition 2.2. *Let f be a minimum Roman dominating function of a graph G without isolated vertices. Let V_i be the sets of vertices v with $f(v) = i$ for $i = 0, 1, 2$. Let $f = (V_0, V_1, V_2)$ be such a function that $|V_1|$ is minimal. Then V_1 is a 2-packing and there is no edge between V_1 and V_2 .*

Clearly, a function $f = (V_0, V_1, V_2)$ is a Roman dominating function if the set V_2 dominates the set V_0 , and the weight of f is $f(V) = \sum_{v \in V} f(v) = 2|V_2| + |V_1|$. Moreover, a Roman dominating function $f = (V_0, V_1, V_2)$ defines a *Roman dominating set* (RDS) $D_R = (V_1, V_2)$, since for each vertex $v \in V$, we have either $v \in V_1 \cup V_2$ or v is dominated by some vertex in V_2 . The weight of a Roman dominating set D_R is $W(D_R) = 2|V_2| + |V_1|$. If D_R is a minimum Roman dominating set of graph G , then $W(D_R) = \gamma_R(G)$. A Roman dominating set D_R is called a *connected Roman dominating set* if it also induces a connected subgraph.

3. Complexity Study

Theorem 3.1. *The minimum Roman dominating set and connected Roman dominating set problems in unit disk graphs are NP-complete.*

Proof. The reduction is from the minimum dominating set problem in grid graphs, which is known to be NP-complete. Given a grid graph $G = (V, E)$, we will construct a unit disk graph $G' = (V', E')$ such that G has a dominating set D with $|D| \leq k$ if and only if G' has a Roman dominating set D_R with $W(D_R) \leq 2k$.

Given any grid graph $G = (V, E)$ with vertex-set $V = \{v_1, v_2, \dots, v_n\}$, where each v_i has integral coordinates (x_i, y_i) and $(v_i, v_j) \in E$ if and only if $\|v_i v_j\| = 1$. For each vertex $v_i \in V$, we add a new vertex u_i with coordinates $(x_i + \frac{1}{10}, y_i)$, and let $U = \{u_1, u_2, \dots, u_n\}$ denote the set associated with V . Now construct the graph $G' = (V', E')$ with vertex-set $V' = V \cup U$, and for any two vertices $u, v \in V'$, $(u, v) \in E'$ if and only if $\|uv\| \leq 1 + 1/10$. It is clear that G' is a unit disk graph, and for each $1 \leq i \leq n$, v_i and u_i have the same neighborhood $N_{G'}(v_i) = N_{G'}(u_i)$, and the same degree $d_{G'}(u_i) = d_{G'}(v_i) = 2d_G(v_i) + 1$. The construction of G' can

clearly be accomplished in polynomial time. Moreover, the following lemmas gives some other properties of graph G' .

Lemma 3.2. *Let $f = (V_0, V_1, V_2)$ be a minimal Roman dominating function of G' . Then $|\{u_i, v_i\} \cap V_2| \leq 1$, for $i = 1, 2, \dots, n$.*

Proof. Suppose, by contradiction, that there exists a j such that $|\{u_j, v_j\} \cap V_2| = 2$. Let $V'_0 = V_0 \cup \{u_j\}$, $V'_1 = V_1$ and $V'_2 = V_2 \setminus \{u_j\}$. It is easy to verify that function $f' = (V'_0, V'_1, V'_2)$ is a Roman dominating function satisfying $f'(V') < f(V')$, a contradicting that f is minimal. $\square$

Lemma 3.3. *Let $f = (V_0, V_1, V_2)$ be a minimal Roman dominating function of G' . Then there exists a minimal Roman dominating function $f' = (V'_0, V'_1, V'_2)$ satisfying $f'(V') = f(V')$ such that $V'_1 = \emptyset$ and $V'_2 \cap U = \emptyset$.*

Proof. Suppose that $V_1 \neq \emptyset$. Then let $v_j \in V_1$ for some j . Thus we have $N_{G'}(v_j) \cap V_2 = \emptyset$ and $N_{G'}(u_j) \cap V_2 = \emptyset$ since $N_{G'}(v_j) = N_{G'}(u_j)$, which implies $u_j \in V_1$. Hence we have either $|\{u_i, v_i\} \cap V_1| = 0$ or 2 for each i . Suppose $V_1 = \{v_{i1}, u_{i1}, v_{i2}, u_{i2}, \dots, v_{ik}, u_{ik}\}$ for some $1 \leq k \leq n$. Let $V'_0 = V_0 \cup \{u_{i1}, u_{i2}, \dots, u_{ik}\}$, $V'_1 = \emptyset$ and $V'_2 = V_2 \cup \{v_{i1}, v_{i2}, \dots, v_{ik}\}$. Clearly, $f' = (V'_0, V'_1, V'_2)$ is a Roman dominating function satisfying $f'(V') = f(V')$.

Now suppose that $V_2 \cap U \neq \emptyset$. Assume, without loss of generality, that $V_2 \cap U = \{u_1, u_2, \dots, u_m\}$ for some $1 \leq m \leq n$. Since $|\{u_i, v_i\} \cap V_2| \leq 1$ for each i , then $V_2 \cap \{v_1, v_2, \dots, v_m\} = \emptyset$. Let $V'_0 = (V_0 \setminus \{v_1, v_2, \dots, v_m\}) \cup \{u_1, u_2, \dots, u_m\}$, and $V'_1 = V_1$ and $V'_2 = (V_2 \setminus \{u_1, u_2, \dots, u_m\}) \cup \{v_1, v_2, \dots, v_m\}$. It is easy to verify that $f' = (V'_0, V'_1, V'_2)$ is a Roman dominating function satisfying $f'(V') = f(V')$ and $V'_2 \cap U = \emptyset$. The lemma is then proved. $\square$

Let D_R be a minimal Roman dominating set of G' . Then by Lemma 3.3, we can assume, without loss of generality, that $D_R = (\emptyset, V_2)$ and $V_2 \cap U = \emptyset$. Suppose that D is a dominating set of G with $|D| \leq k$. Let $D_R = (\emptyset, D)$, then D_R is a Roman dominating set with $W(D_R) = 2|D| \leq 2k$. Conversely, suppose that G' has a Roman dominating set $D_R = (\emptyset, V_2)$, $V_2 \cap U = \emptyset$ and $W(D_R) = 2|V_2| \leq 2k$. Then $V_2 \subseteq V$ is also a dominating set of G , $|V_2| = \frac{1}{2}W(D_R) \leq k$. Let $D = V_2$, then G has a dominating set D with $|D| \leq k$. Therefore, G has a dominating set D with $|D| \leq k$ if and only if G' has a Roman dominating set D_R with $W(D_R) \leq 2k$. Thus, the minimum Roman dominating set problem in unit disk graph is NP-complete.

In addition, D is a connected dominating set in G if and only if $D_R = (\emptyset, D)$ is a connected Roman dominating set in G' , and the minimum connected dominating set problem in grid graph is NP-complete. Thus, the connected Roman dominating set problem in unit disk graphs is also NP-complete. $\square$

4. Approximation Algorithms

4.1. Algorithm for computing minimum RDS

It is well known that any maximal independent set D is also a dominating set. Thus $D_R = (\emptyset, D)$ is a Roman dominating set. One method of finding a Roman dominating set in a graph is to find a maximal independent set. A straightforward method of finding a maximal independent set is as follows: Select an arbitrary vertex $v \in V$, add v to the current independent set D , which is initially set to be an empty set, and delete $N_G(v)$ from the current graph G , and then repeat the process until all vertices in V have been deleted.

Algorithm A 5-Approximation Algorithm for MRDS

1. $S := V, D := \emptyset$
 2. **while** $S \neq \emptyset$ **do**
 3. Choose a node $v \in S$
 4. $D := D \cup \{v\}, S := S \setminus N_G(v)$
 5. **end while**
 6. **return** $D_R := (\emptyset, D)$
-

Theorem 4.1. *Let $G = (V, E)$ be a unit disk graph, Algorithm A returns a Roman dominating set in time of $O(|E|)$ whose size is at most five times that of the minimum Roman dominating set.*

Proof. It is clear that the algorithm returns a Roman dominating set in time of $O(|E|)$. Let $D_R^* = (V_1^*, V_2^*)$ denote an optimal Roman dominating set of G such that V_1^* is a minimal. By Proposition 2.2 we have V_1^* as a 2-packing and there is no edge between V_1^* and V_2^* . Note that G is a unit disk graph and the returned set D is an independent set. Moreover, no vertex in V_2^* can dominate more than 5 vertices in $D \setminus V_1^*$, and each vertex in V_1^* only can dominate itself. Hence we have

$$|D| - |V_1^* \cap D| \leq 5|V_2^*|,$$

and

$$W(D_R) = 2|D| \leq 5(2|V_2^*| + |V_1^*|) = 5\gamma_R(G)$$

and then the theorem follows. $\square$

4.2. Algorithm for computing minimum connected RDS

Lots of efforts have been made to design approximation algorithms for minimum connected dominating set problem. Wan *et al.* [10] proposed a two-phase distributed algorithm for the problem in UDGs. The algorithm first constructs a spanning tree, and then at the first phase, each node in a tree is examined to find a maximal independent set I and all the nodes in the MIS are colored black. At the second

phase, more nodes are added (color blue) to connect those black nodes. Let C be the nodes adding in the second phase, and $|C| < |I|$.

Our algorithm for computing the minimum connected RDS in UDG is first to construct a connected dominating set $S = I \cup C$ using the algorithm in [10]. Let $V_1 = C$ and $V_2 = I$. Then $D_R = (V_1, V_2)$ is a connected RDS.

Algorithm B for computing minimum connected RDS

1. Construct a CDS $S = I \cup C$ using the algorithm in [10]
 2. $V_1 = C$, $V_2 = I$
 3. **return** $D_R = (V_1, V_2)$
-

Theorem 4.2. *Let $G = (V, E)$ be a unit disk graph, Algorithm B returns a connected Roman dominating set with approximation ratio 7.5.*

Proof. Any maximal independent set is a dominating set. Then V_2 is a DS. By the first step of Algorithm B, $V_1 \cup V_2$ is a CDS and thus $V_1 \cup V_2$ is connected. So, $D_R = (V_1, V_2)$ is a connected Roman dominating set of G . Let $D_R^* = (V_1^*, V_2^*)$ denote an optimal connected Roman dominating set of G . By Theorem 4.1, $2|V_2| \leq 5W(D_R^*)$. Moreover, $|V_1| < |V_2|$. Hence we have

$$W(D_R) = 2|V_2| + |V_1| < 7.5W(D_R^*),$$

and then the theorem follows. □

5. Conclusion

In this paper we first show that the Roman dominating set and connected Roman dominating set problems in unit disk graphs are NP-complete, and then present two approximation algorithms for these problems.

Grid graphs compose a subclass of unit disk graphs. However, hardness results for the Roman domination problem in grid graphs are still unknown (to the best knowledge of the authors). This raises an interesting problem for future work.

References

- [1] B. N. Clark, C. J. Colburn and D. S. Johnson, Unit disk graphs, *Discrete Math.* **86** (1990) 165–177.
- [2] E. J. Cockayne, P. A. Dreyer, S. M. Hedetniemi and S. T. Hedetniemi, Roman domination in graphs, *Discrete Math.* **278** (2004) 11–22.
- [3] P. A. Dreyer, Applications and variations of domination in graphs, PhD thesis, Rutgers University, New Jersey (2000).
- [4] M. A. Henning, A characterization of Roman trees, *Discuss. Math. Graph Theory* **22** (2002) 325–334.
- [5] M. A. Henning and S. T. Hedetniemi, Defending the Roman empire — a new strategy, *Discrete Math.* **266** (2003) 239–251.

- [6] H. B. Hunt III, M. V. Marathe, V. Radhakrishnan, S. S. Ravi, D. J. Rosenkrantz and R. E. Stearns, NC-approximation schemes for NP-and PSPACE-hard problems for geometric graphs, *J. Algorithms* **26** (1998) 238–274.
- [7] M. V. Marathe, N. Breu, H. B. Hunt III, S. S. Ravi and D. J. Rosenkrantz, Simple heuristics for unit disk graphs, *Networks* **25** (1995) 59–68.
- [8] A. Pagourtzis, P. Penna, K. Schlude, K. Steinhofel, D. Taylor and P. Widmayer, Server placements, Roman domination and other dominating set variants, in *Proc. Second International Conference on Theoretical Computer Science* (2002), pp. 280–291.
- [9] I. Stewart, Defend the Roman Empire! *Sci. American*, December (1999) 136–138.
- [10] P.-J. Wan, K. M. Alzoubi and O. Frieder, Distributed construction of connected dominating set in wireless ad hoc networks, *Mobile Net. Appl.* **9** (2004) 141–149.